

Groups and Lie Algebras: Exercise Sheet 1

1. The symmetric group S_n

We have defined the symmetric group S_n as the group of permutations on the set $\{1, 2, \dots, n\}$. A permutation σ acts on elements $i \in \{1, 2, \dots, n\}$ by

$$i \mapsto \sigma(i).$$

We also denote this permutation as

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}.$$

For $1 \leq k \leq n$, let $a_1, a_2, \dots, a_k \in \{1, 2, \dots, n\}$ be pairwise different. A *cycle* $(a_1, \dots, a_k)$ is the cyclic permutation $a_1 \mapsto a_2, a_2 \mapsto a_3, \dots, a_k \mapsto a_1$; it acts as the identity on all other elements. We call k the length of the cycle.

- a) Show that any permutation σ can be written as a product of disjoint cycles (two cycles are disjoint if they contain no common elements).
- b) A *transposition* is a cycle of length 2. For $n \geq 2$, show that every cycle (and thus every permutation) can be written as a product of transpositions. This means that S_n is *generated* by transpositions.
- c) Give an example that shows that the decomposition of a cycle into transpositions is not necessarily unique.
- d) Denote by τ_i the cycle $(i, i+1)$ for $1 \leq i < n$. Show that they satisfy the relations

$$\tau_i^2 = 1, \quad \tau_{i+1}\tau_i\tau_{i+1} = \tau_i\tau_{i+1}\tau_i, \text{ and } \tau_i\tau_j = \tau_j\tau_i \text{ for } 2 \geq |i-j|.$$

- e) If σ is decomposed into disjoint cycles of length k_i ($i \leq n$), define the *sign* of σ as

$$\text{sign}(\sigma) = (-1)^{N(\sigma)}, \quad \text{where } N(\sigma) = \sum_i (k_i - 1).$$

Using the fact that two products of τ_i represent the same permutation if and only if they are related by the relations from d), show that sign is a group homomorphism from S_n to $(\{\pm 1\}, \cdot)$.

2. Chinese remainder theorem

Show that if a and b are coprime positive integers, then $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ is isomorphic to $\mathbb{Z}/ab\mathbb{Z}$. How can you use this result to decompose $\mathbb{Z}/n\mathbb{Z}$ for an arbitrary $n \in \mathbb{N}$?

3. **Automorphisms of $\mathbb{Z}_n$** Let G be a group. Show that the set $\text{Aut}(G)$ of invertible group homomorphisms $G \rightarrow G$ is a group. For $n \in \mathbb{N}$ compute the group $\text{Aut}(\mathbb{Z}_n)$.

4. **The Lorentz group** Recall that the Lorentz group $O(3, 1)$ is the group of linear transformations $\Lambda: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ (in special relativity the basis of $\mathbb{R}^4$ is usually denoted by (e_0, e_1, e_2, e_3)) that preserve the inner product $-x_0^2 + x_1^2 + x_2^2 + x_3^2$. We say that $\Lambda \in O(3, 1)$ is *orthochronous* if the first component of $\Lambda(e_0)$ is positive. Otherwise, we say that it is *time reversing*. $\Lambda \in O(3, 1)$ is *parity reversing* if it reverses orientation and is orthochronous.

- a) Show that there is a group homomorphism $F: O(3, 1) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$ such that the time and parity reversing elements are sent to $(1, 0)$ and $(0, 1)$, respectively.
- b) Show that F has a section, i.e. that there exists a group homomorphism $s: \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow O(3, 1)$ such that $F \circ s = \text{id}$
- c) The kernel of F is called the *proper orthochronous Lorentz group* $SO(3, 1)^+$. Work out, recall, or look up a description of the elements of $SO(3, 1)^+$ in terms of boosts and rotations. Show that $O(3, 1) \cong SO(3, 1)^+ \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$ as sets (not necessarily as groups). Is this also an isomorphism of groups?
- d) Show that conjugation by $s(-)$ induces a group homomorphism $\varphi: \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \text{Aut}(SO(3, 1)^+)$ from $\mathbb{Z}_2 \times \mathbb{Z}_2$ to the group $\text{Aut}(SO(3, 1)^+)$ of automorphisms of $SO(3, 1)^+$.
- e) Equip the set $SO(3, 1)^+ \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$ with the group structure:

$$(\Lambda, (a, b)) \cdot (\Lambda', (a', b')) = (\Lambda \varphi(a, b)(\Lambda'), (a + a', b + b')).$$

Show that this group is isomorphic to $O(3, 1)$.